UNIVERSITÀ DEGLI STUDI DI PADOVA

Introducing CMake

Stefano Ghidoni

DIPARTIMENTO DI INGEGNERIA DELL'INFORMAZIONE

INTELLIGENT AUTONOMOUS SYSTEMS LAB

Agenda

- Describing a software project
- Building systems
- CMake
  - Advantages
  - User's perspective
  - Developer's perspective

Describing a project

- A software project is usually composed of several source files (.h, .hpp, .cpp)
- Complex projects are composed of:
  - One or more executable(s)
  - One or more library/ies

Describing a project

- Source files alone do not provide a description of the project
- Need to know the structure of the SW project
  - Deeper understanding of the SW project
  - Info about how to compile

Describing a project

- A SW project can be described by means of its **targets**
- Compilation targets
  - Create an executable
  - Create a library
- In both cases
  - List of the .cpp files to be compiled together
  - List of the libraries to be linked

Describing a project

- How can we describe a project?
- Description based on **compile command**
  - Impractical
  - Hard to interpret
- Description based on **IDE project**
  - Convenient
  - Platform-dependent
- Description based on a **building system**

Building systems

- A building system handles the compilation process
- Several options
  - Make, Ninja, …
  - Sometimes depend on the platform

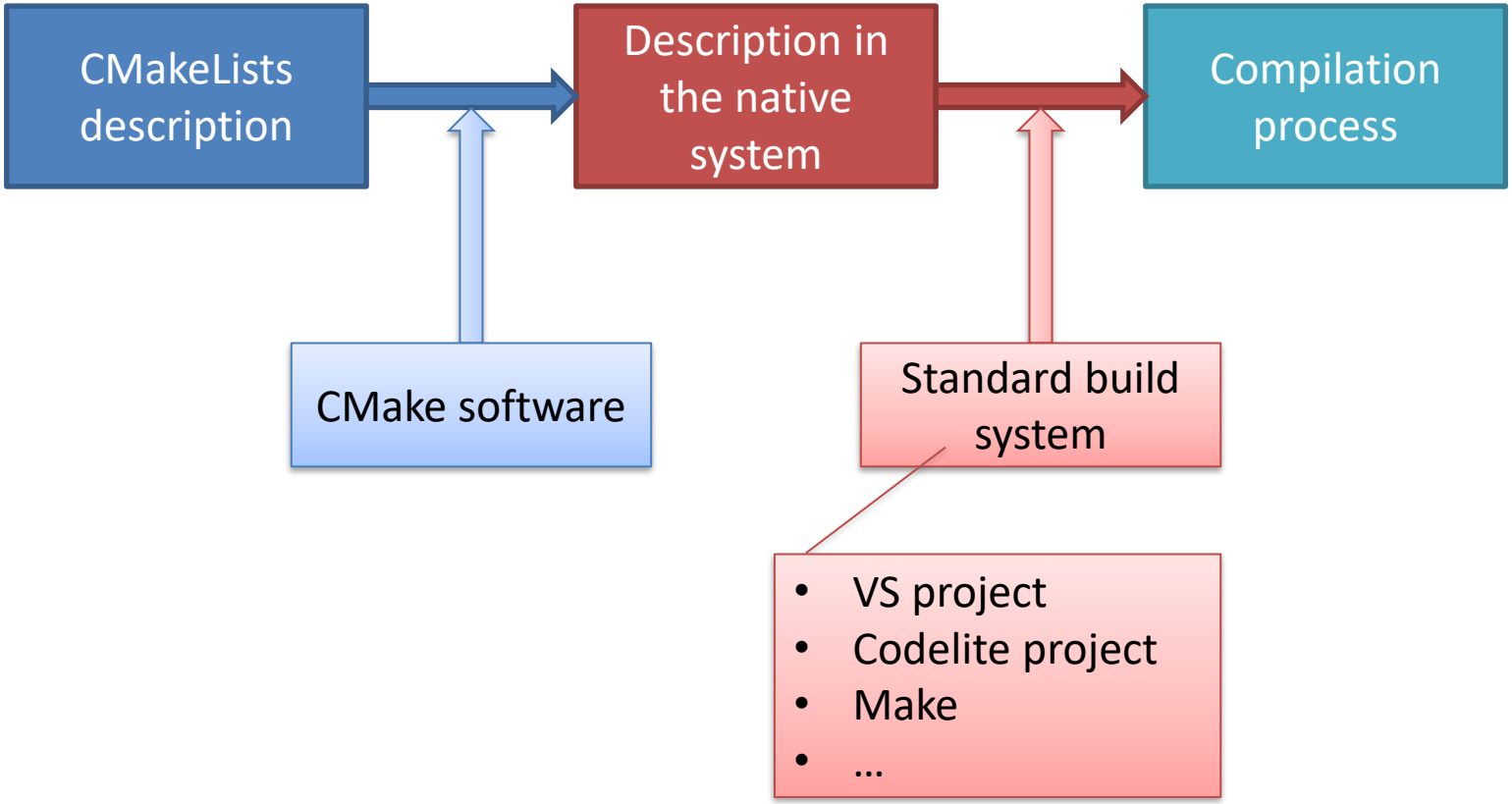
Introducing CMake

- CMake is a standard meta-building system
- Independent from:
  - Compiler
  - Platform
- Used in combination with the "local" building system
- Supported by many IDEs

Introducing CMake

- CMake can be seen as a way for describing projects
- It defines:
  - Compilation targets
    - Executables
    - Libraries
  - Libraries to be linked
- Based on text files named CMakeLists.txt

CMake: process

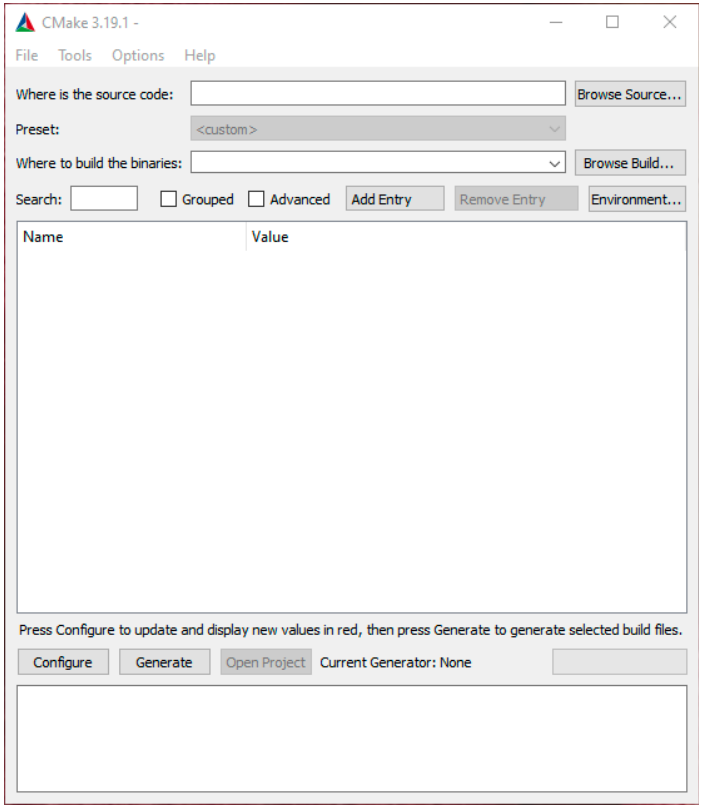


CMake homepage

User's perspective

CMake software

- GUI or command-line
- Two activities:
  - Configure
  - Generate



User's perspective

- Typical workflow:
  - **Call CMake** (GUI/command line)
  - Select the directory containing the source
  - Select the directory where the compiler output shall go
    - Standard: [project root]/build
  - **Configure**
  - (possible) module enable/disable
  - **Generate**

Compiling the project

- The *generate* action creates the files to be provided to the build system
  - E.g.: VS solution
- From that point, the compilation process gets out of CMake's scope

Compiling the project

- Example under Linux

```
mkdir build
cd build
cmake ..
make
```

Compilation results in a separate folder (optional)

Compiling the project

- Example under Linux

```
mkdir build
cd build
cmake ..
make
```

- Generates compiler specific files
- **No compilation done yet!**

Compiling the project

- Example under Linux

```
mkdir build
cd build
cmake ..
make
```

- Compiles everything using a Makefile (generated by CMake)

Developer's perspective

Developer's perspective

- Describe the project using the CMake standard
  - Create CMakeLists.txt file manually
  - Use an IDE to generate CMakeLists.txt automatically
    - More complex
    - Do not modify them directly – only through the IDE

CMakeLists.txt

- Choose a minimum CMake version
  - 2.8 is usually enough
    - **Do not use more recent versions** if you do not need advanced features
      - This is a **minimum requirement**

Sets the mimimum required version of CMake.

```
cmake_minimum_required(VERSION 3.2 FATAL_ERROR)
```

CMakeLists.txt

- Give a name to the project

Set the name and version; enable languages.

```
project(<name> VERSION <version> LANGUAGES CXX)
```

- CMake sets several variables based on project().
- Call to project() must be direct, not through a function/macro/include.
- CMake will add a call to project() if not found on the top level.

CMakeLists.txt

- List libraries needed by the project
  - E.g., OpenCV

Finds preinstalled dependencies

```
find_package(Qt5 REQUIRED COMPONENTS Widgets)
```

- Can set some variables and define imported targets.
- Supports components.

CMakeLists.txt

- **Target:** creating an executable
  - Corresponds to: g++ -o tool –c main.cpp another_file.cpp

Add an executable target.

```
add_executable(tool
  main.cpp
  another_file.cpp
  )
```

CMakeLists.txt

- **Target:** creating a library

Add a library target.

```
add_library(foo STATIC
  foo1.cpp
  foo2.cpp
  )
```

- Libraries can be STATIC, SHARED, MODULE, or INTERFACE.
- Default can be controlled with BUILD_SHARED_LIBS.

CMake – a simple example


```
cmake_minimum_required(VERSION 3.2 FATAL_ERROR)

project (tuple)

include_directories(include)

add_library(tuple
    include/tuple.h
    src/tuple.cpp
)

add_executable(add_tuple
    src/add_tuple.cpp
)
target_link_libraries(add_tuple
    tuple
)
```

CMake – working with OpenCV


```
cmake_minimum_required(VERSION 2.8)
project(test1)

find_package(OpenCV REQUIRED)

include_directories(${OpenCV_INCLUDE_DIRS})

add_executable(${PROJECT_NAME} src/test1.cpp)
target_link_libraries(${PROJECT_NAME} ${OpenCV_LIBS})
```

CMake – working with OpenCV


```
1  cmake_minimum_required(VERSION 2.8)
2  project(test1)
3
4  find_package(OpenCV REQUIRED)
5
6  include_directories(${OpenCV_INCLUDE_DIRS})
7
8  add_executable(${PROJECT_NAME} src/test1.cpp)
9  target_link_libraries(${PROJECT_NAME} ${OpenCV_LIBS})
```

Looks for OpenCVConfig.cmake (where a bunch of OpenCV related vars are defined)

CMake – working with OpenCV


```
cmake_minimum_required(VERSION 2.8)
project(test1)

find_package(OpenCV REQUIRED)

include_directories(${OpenCV_INCLUDE_DIRS})

add_executable(${PROJECT_NAME} src/test1.cpp)
target_link_libraries(${PROJECT_NAME} ${OpenCV_LIBS})
```

Tells the compiler to look for the headers also in the folders defined in this variable

CMake – working with OpenCV


```
1  cmake_minimum_required(VERSION 2.8)
2  project(test1)
3
4  find_package(OpenCV REQUIRED)
5
6  include_directories(${OpenCV_INCLUDE_DIRS})
7
8  add_executable(${PROJECT_NAME} src/test1.cpp)
9  target_link_libraries(${PROJECT_NAME} ${OpenCV_LIBS})
```

Tells the compiler to build an executable named test1, the source file for the compilation is just src/test1.cpp

CMake – working with OpenCV


```
1  cmake_minimum_required(VERSION 2.8)
2  project(test1)
3
4  find_package(OpenCV REQUIRED)
5
6  include_directories(${OpenCV_INCLUDE_DIRS})
7
8  add_executable(${PROJECT_NAME} src/test1.cpp)
9  target_link_libraries(${PROJECT_NAME} ${OpenCV_LIBS})
```

Tells the linker to to link the test1 executable with the libraries defined by the variable

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